MILITARY SPECIFICATION

PENETRATOR, CORE AND NOSE COMPONENTS FOR PROJECTILE, 105MM, APDS-T, M392A1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers components made of high density tungsten for use as penetrators on 105MM kinetic energy tank ammunition. This specification is intended to be a supplement to MIL-P-50370 and components made to this specification shall also comply with the requirements contained in MIL-P-50370 and applicable drawings.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-P-50370	- Projectile, 105MM, APDS-T, M392A1 Metal Parts Assembly

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

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2.1 (Cont'd)

STANDARDS (Cont'd)

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-1169	- Packaging, Packing and Marking for Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

C8595471	- Core
C8595472	- Nose (Machined) Sheet 1 of 2, Nose, Blank, Sheet 2 of 2.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other Publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 - Surface Texture

(Application for copies shall be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

ASTM-B311	- Standard Method of Test for Density of Cemented Carbides
ASTM-E-8	- Standard Method of Tension Testing of Metallic Materials
ASTM-E18	- Standard Method of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies of the above publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Phila. Pa. 19103).

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3. REQUIREMENTS

3.1 General.-The components shall be manufactured by the powder metallurgy process and shall comply with the requirements on the applicable drawings, the referenced specification and the following:

3.2 Chemical Properties.

3.2.1 Core.-The following chemical composition requirements shall apply to each core powder batch after milling and to each core component. The percent by weight of tungsten carbide shall be 87.0% to 89.0%; the percent by weight of cobalt shall be 11.0% to 13.0%. Samples selected from any batch of mixed powder shall not contain more than 0.7% iron.

3.2.2 Nose.-The following chemical composition requirements shall apply to each nose powder blend lot and each nose component. The minimum percent by weight of tungsten shall be 89.0%. The balance of the composition is advisory 7.5% Nickel, 2.5% Copper, and 1.0% other elements.

3.3 Mechanical Properties.

3.3.1 Ultimate Tensile Strength.-Nose components shall have a minimum ultimate tensile strength of 115,000 pounds per square inch (psi).

3.3.2 Elongation.-Nose components shall have a minimum elongation of 10.0%.

3.3.3 Compression.-Nose component compression test specimens shall show no sign of cracking when compressed to one-half of the orignial height. A compressive yield strength of 120,000 pounds per square inch (psi) at 1.0% offset is advisory.

3.3.4 Hardness.

3.3.4.1 Core.-The core components shall have a hardness value within the range of 86-89 on the Rockwell "A" scale.

3.3.4.2 Nose.-The nose components shall have a hardness value within the range of 28-32 on the Rockwell "C" scale.

3.4 Physical Properties.

3.4.1 Core.-The core components shall have a minimum density of 14.10 grams per cubic centimeter (gm/cc).

3.4.2 Nose.-The nose components shall have a minimum density of 16.90 grams per cubic centimeter (gm/cc).

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3.5 Material Defects.-All components shall be free from cracks, excessive porosity and any other similar defect.

3.6 Burrs.-No components shall have a burr which might interfere with assembly or functioning of the projectile or which might be injurious to personnel handling the item.

3.7 Foreign Matter.-All components shall be free of chips, dirt, grease, solvent, rust corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.1.3 Inspection Criteria.

a. Successful passage of all inspections required by the drawings and specifications shall be considered as a condition precedent to performing the proving ground function tests.

b. Acceptance or rejection of a lot by the Government shall be considered as a condition precedent to performing the proving ground functional tests.

4.2 First Article Sample.

4.2.1 Initial Production Sample.-At the beginning of regular production, a sample shall be submitted in accordance with the contract requirements and shall consist of five (5) each nose and core components, and five (5) each as-sintered nose blanks.

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4.2.1.1 Examination and Test.-After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial Production Sample Failure.-Failure of the sample to comply with the requirements of the drawings and specification shall result in sample disapproval.

4.3 Inspection Provisions.

4.3.1 Lot.-A lot of cores or noses shall consist of not more than one blended powder batch which has been processed under the same conditions.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used except where otherwise specified. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for roughness determination.

4.3.2.1 Core Drawing C8595471.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Weight	Approved Scale
102.	Hardness	Hardness Tester
103.	Runout of tip with body	Gage
104.	Runout of basic taper with body	Gage
105.	Perpendicularity of rear face with body	Gage
106.	Diameter of body	Gage
107.	Length to small basic diameter	Gage
108.	Length to large basic diameter	Gage
109.	Straightness	Gage
110.	Angle of tip	Gage
111.	Basic Taper	Gage
112.	Metal Defective (see Sketch 1)	Visual

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4.3.2.1 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Diameter of chamfer	Gage
202.	Angle of base	Visual (see 6.2)
203.	Maximum flat of nose (if applicable)	Visual (see 6.2)
204.	Depth of recess in rear face (if applicable)	Visual (see 6.2)
205.	Surface finish, improper	Visual (see 6.2)
206.	Burr	Visual
207.	Foreign matter	Visual

4.3.2.2 Nose (Machined), Drawing 8595472.

CRITICAL		
	None defined	
MAJOR		
101.	Hardness	Hardness Tester
102.	Runout of tip with diameter forward of flange	Gage
103.	Runout of cavity with diameter forward of flange	Gage
104.	True position of flange with diameter forward of flange	Gage
105.	Diameter forward of flange	Gage
106.	Diameter of flange	Gage
107.	Depth of cavity	Gage
108.	Profile of cavity	Gage
109.	Thickness of flange	Gage
110.	Metal defective	Visual
MINOR		
201.	Length from base to small basic diameter	Gage
202.	Length from base to large basic diameter	Gage
203.	Length, total	Gage
204.	Radii or chamfer, missing or incorrect	Visual (see 6.2)
205.	Surface finish, improper	Visual
206.	Burr	Visual
207.	Foreign matter	Visual

4.3.3 Inspection Testing Methods and Procedures.-The following tests ahall be performed in accordance with the specified procedures.

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4.3.3.1 Chemical Analysis.-The chemical analysis determination shall be by methods and procedures subject to the approval of the Procuring Contracting Officer. Certification of compliance with the respective chemistry requirements for the core and nose shall accompany each core and nose lot.

4.3.3.2 Mechanical Properties.

4.3.3.2.1 Ultimate Tensile Strength and Elongation.-Four nose blanks shall be randomly selected from each nose lot and tested for compliance with the requirements of paragraphs 3.3.1 and 3.3.2. From each nose blank selected, one longitudinal and one transverse tension test specimen shall be prepared and tested. Failure of any test specimen shall be cause for rejection of the lot represented by the failed test specimen. The location of the test specimens shall be in accordance with sketch #2. The tension test specimen shall be in conformance to the dimensions in sketch #3. Care must be exercised in preparing the tension test specimen in terms of surface finish, end parallelism, and blend radii in order to obtain reproducible test results. The specimen shall be tested in a tensile test machine with a constant cross head speed of 0.015 inch per minute. The total time of the test shall not exceed twelve (12) minutes. The ultimate tensile strength shall be determined by dividing the load at fracture by the original diameter of the gage length of the specimen measured to an accuracy of $\pm$ 0.001 inch. The total length of the specimen shall be measured prior to testing to an accuracy of 0.0005 inch. After testing the broken specimen shall be fitted back together and the total length again measured to 0.0005 inch accuracy. The percent elongation shall be determined by the following:

$$\% \text{ Elongation} = \frac{(\text{Length after Fracture} - \text{Original Length}) \times 100}{(\text{Original Gage Length})}$$

4.3.3.2.2 Compression Test.-From each nose blank selected for the tension test, one compression test specimen shall be prepared and tested. Failure of any test specimen to comply with the requirements of paragraph 3.3.3 shall be cause for rejection of the lot represented by the failed test specimen. The location of the test specimens shall be 0.357 $\pm$ 0.005 inch in diameter by 0.500 $\pm$ 0.0005 inch in length. The ends of the test shall be parallel within 0.001 inch. Care must be taken in the preparation of the test specimen so as not to influence test results. A machined finish on the cylindrical surface of 64 micro inches or better is acceptable. The ends of the specimen shall be ground to a finish of 16 micro inches or better. The specimen shall be placed on the bed of a tensile testing machine and a load applied such that a reduction in height of the specimen of 50% shall be attained while under load. Positive stops or "jo" blocks may be used, provided that the specimen is prepared with sufficient percision to guarantee that contact with the stop represents 50% compression. A deflectometer shall be connected with the moving member of the machine such that a strain diagram may be recorded and the compressive yield strength of 1.0% offset, determined. Rate of strain during test shall be from 0.05 to 0.10 inches per minute. Upon completion of pressing, the specimen shall be removed and examined visually for signs of cracking.

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4.3.3.2.3 After 10,000 consecutive noses have been accepted for compliance with the tensile, elongation and compressive requirements of paragraphs 3.3.1, 3.3.2 and 3.3.3, the sampling may be reduced to two noses per lot for tensile and compression tests, upon the approval of the procuring contracting officer. If a lot of noses is rejected under 4.3.3.2.1 or 4.3.3.2.2, sampling shall revert to that originally specified until the requirements for reduced sampling are again met.

4.3.3.2.4 Retest.-If only one nose fails to comply with tensile, elongation and compression requirements of paragraphs 3.3.1, 3.3.2 and 3.3.3, four additional noses shall be selected from the failed lot and tested for the characteristic which failed. If all the retest pieces pass this test, the lot shall be accepted. If one or more fail this test, the lot represented by the failed samples will be rejected.

4.3.3.2.5 Hardness.-Samples of cores and noses selected in accordance with paragraphs 4.3.2.1 and 4.3.2.2 shall be subjected to this test. Hardness values on these components shall be determined utilizing the equipment and procedures established in ASTM specification E18, using the "A" scale for cores and the "C" scale for noses. Failure of any one hardness reading to meet the requirements of paragraph 3.3.4 will result in rejection of the component with no retest allowed. Failure of a sample group to comply with the acceptance criteria of MIL-STD-105 will result in rejection of the lot represneted by the sample.

4.3.3.2.5.1 Location.

4.3.3.2.5.1.1 Core.-Four hardness tests shall be taken on the cylindrical surface of each core tested. Tests shall be taken along the length, at quarter of an inch intervals, from the rear face, and shall be spaced 90 degrees from each other around the circumference.

4.3.3.2.5.1.2 Nose.-Two hardness tests shall be taken on the rear face of the nose at mid-width, ninety degrees apart.

4.3.3.2.5.2 Calibration.-Hardness test equipment shall be calibrated every four hours during continuous production and at the start of production following a shutdown in excess of four hours.

4.3.3.3 Physical Properties.

4.3.3.3.1 Density, core and nose.-Density determination shall be made by a test method approved by the responsible technical agency. The procedure shall incorporate a balance with an accuracy of at least 0.010 grams. In the event of any dispute, the procedure established by ASTM specification B311 shall be used as a reference. Four cones and four noses shall be randomly selected from each lot of cores and noses and subjected to the approved test method. Failures of any component to comply with the requirements of paragraph 3.3.4 will result in rejection of the lot represented by the component.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking.

5.1.1 Level A - Military package - Not applicable (NA)

5.1.2 Level B - Limited military package - N/A

5.1.3 Level C.- Minimum military package - Preservation, packaging, packing, and marking shall comply with the applicable drawing, specifications, Military Specification MIL-STD-1169 and this specification.

5.2 Packing.-Core and Nose components shall be packed in a manner which will insure acceptance by common carrier and will afford protection against physical damage during shipment from the supply source to the first receiving activity for immediate use. Shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.3 Marking.-In addition to any special marking required by the contract, or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot must be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Component Identity.-It is recommended that the component identity be maintained, starting with the powder blend lot, through to and including final machining. The component marking should include manufacture's symbols, and powder blend lot.

6.2 Visual Examination Qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

Custodian:

Army - MU

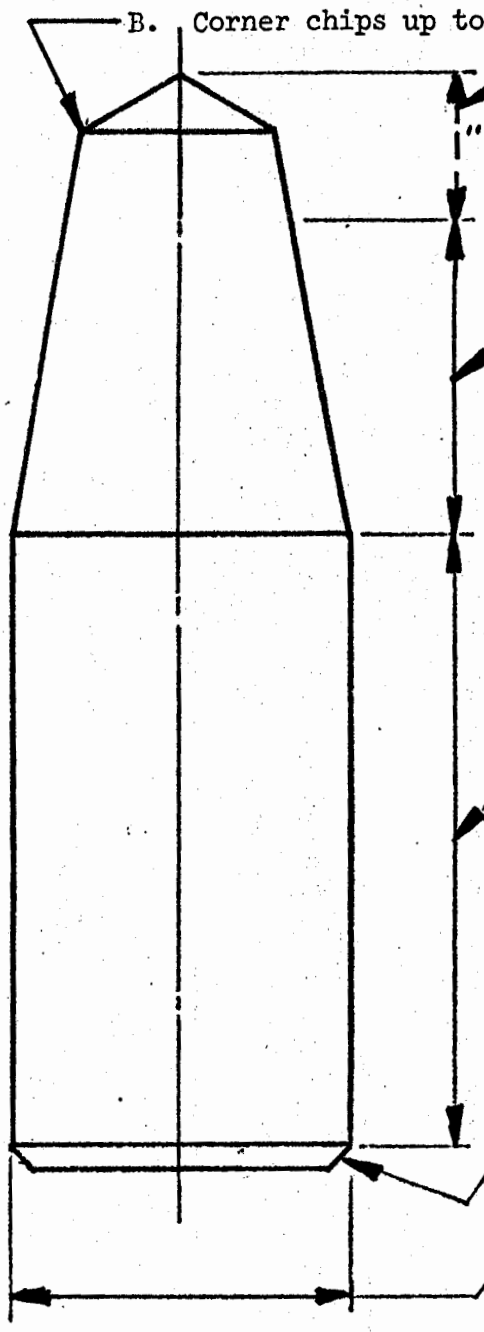
Preparing activity:

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Project no. 1315-A252

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VISUAL STANDARDS FOR CORE, M392A1
METAL DEFECTS



B. Corner chips up to 1/32" dia. (showing no raised metal) acceptable

A. No holes, or evidence of porosity permissible.

C. Depressions, or surface imperfections up to 1/32" dia. acceptable where bottom is rounded and can be clearly seen.

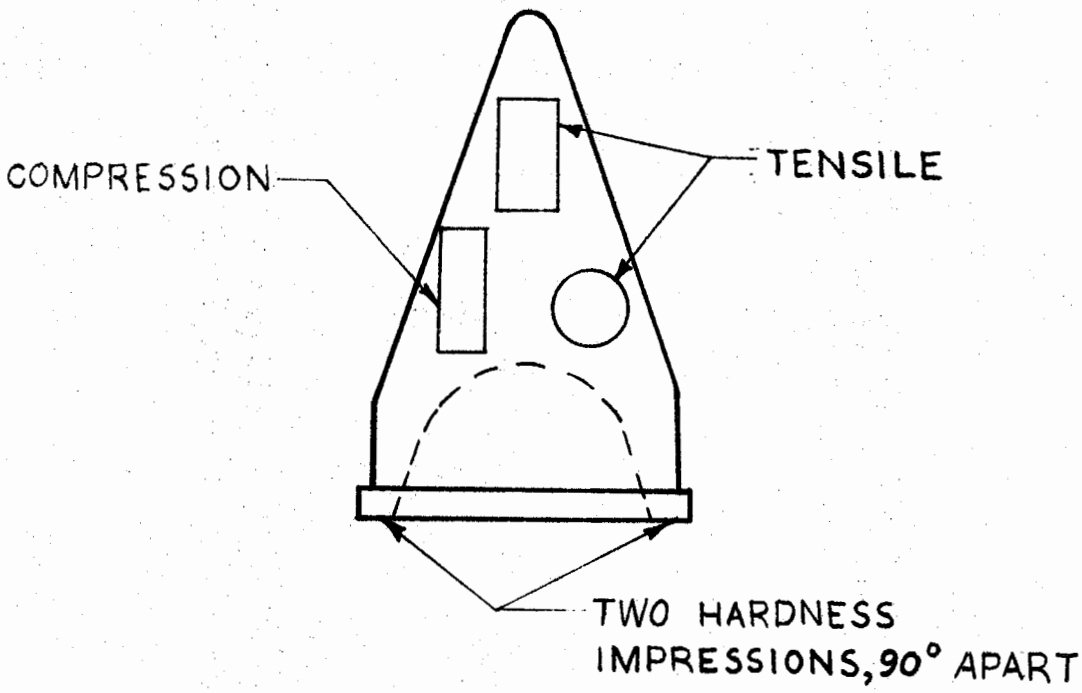
D. Same as C., except that both C and D are not allowed on a single core. Total area of imperfections not to exceed 2 square inches.

E. Same as B.

F. Ground Surface (32 finish) required edge to edge. Evidence of porosity or defects same as C & D. Acceptable, as long as defect occurs in no other location.

SKETCH #1

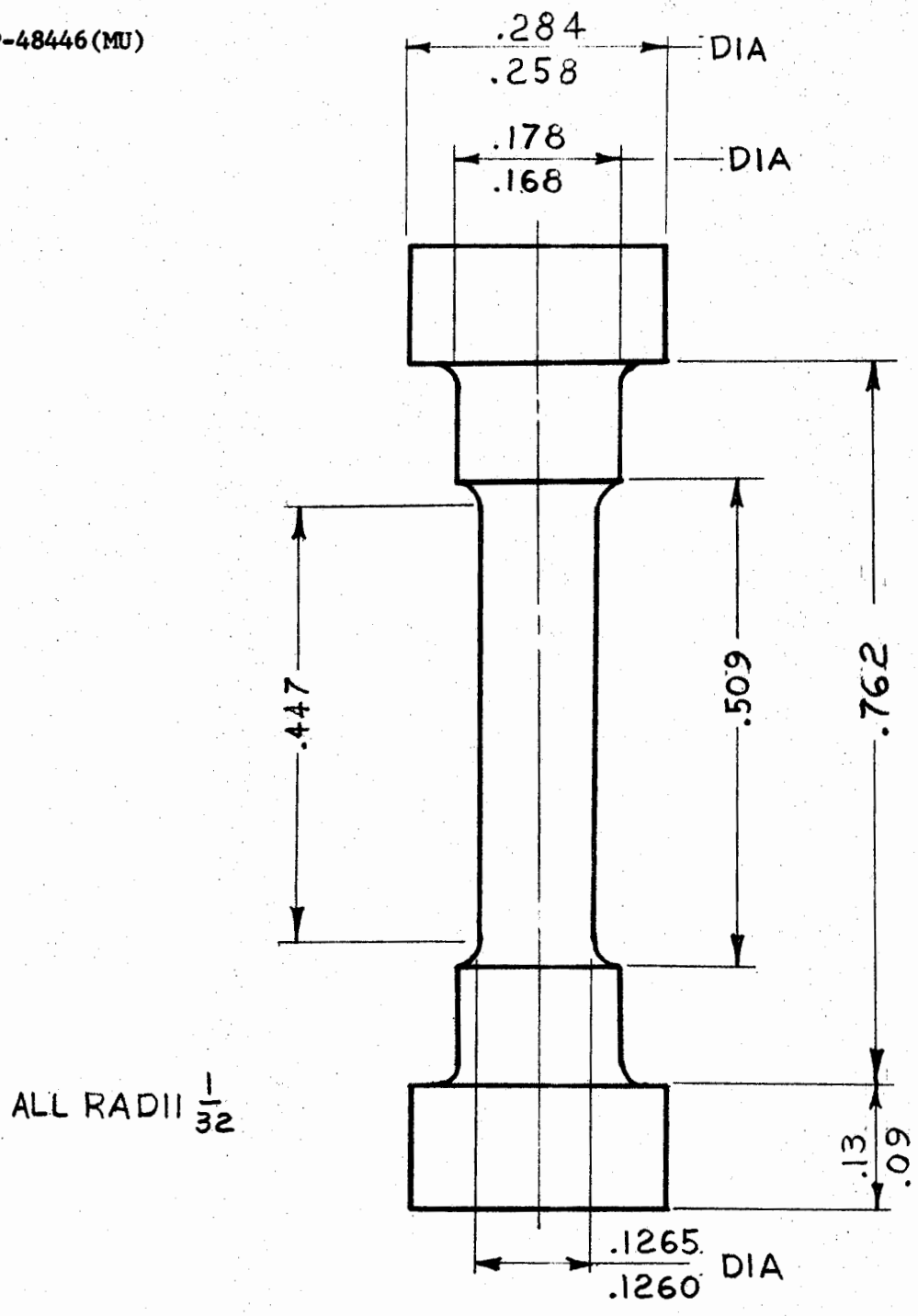
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LOCATION OF HARDNESS TESTS, TENSION AND
COMPRESSION TEST SPECIMENS FROM NOSE COMPONENT

SKETCH #2

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TENSION TEST SPECIMEN

SKETCH #3

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